

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	TAS A
People present	ASO, ARU
Purpose	Maintenance installation / take-down / quick visit / emergency
Date	10/8-23

Notes: MAGNETIC + 21° = TRUE NORTH

NEW LOGGER PROGRAM AFTER DOWNLOAD : PROMICE_V2-2022-V1-2.CRI

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	N			Clean? Y / N
Ablation hose assembly	N	N			No air in hose and bladder half full? (Y) N
X Iridium modem IMEI number	Y (N)	used - N	Record this number!!	300 234 0 6631 6280	
X Inclinator/compass	Y	Y	Attached to the radiometer? Y (N)		
Satellite antenna	N	N			
Wind sensor	N	N	Black box pointing toward the radiometer (i.e., along the boom)? (Y) N		
X Temperature / humidity assembly	Y (rect)	Y	Fan spins? (Y) N The shipped one gave 0 humidity Sounds, OK? (Y) N and 0 temp, so had to replace with old one.		
HygroClip T/hum assembly	N	N	NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
X Sonic ranger on AWS	Y	Y	Old sensor with new membrane? Y (N)		
X Sonic ranger on stake(s)	Y	Y	Old sensor with new membrane? Y (N) NB: make sure not being buried next winter		
X Thermistor string	Y	N			
Solar panel	N	N	Clean? (Y) N NB: is not effective when partially covered		
GPS antenna	N	Y	At top of logger enclosure? (Y) N No velcro in new box, so does not stick		
X Logger box ID	Y (T2-0004)	T2-0003	CR1000 or CR1000X? ID number: T2-0003		
Multiplexer	N	N	Clicking during measurement cycle? (Y) N		
X Logger enclosure	Y	Y	Replaced (including everything inside)? (Y) N Put in new GPS. New desiccant bags? (Y) N		
Battery box	N	N	New box? Y (N) New desiccant bags? (Y) N		
X Stakes	Y	N	New one(s) drilled in? Y (N) Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N	N	New tripod? Y (N) New sensor boom? Y (N)		

Station name	TAS A
Date	10 August 2023

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	+1m:13s	0
Download data and/or change CF/SD Card (wait for green light)	took CF card Y/N	Y/N
Name of logger program	2015	Prohila V2-2022-V1-2
Screen dump of values in fast scan mode	(Y/N)	(Y/N)
Photos of tripod, sensors, logger box wiring etc. (more is better)	* (Y/N)	(Y/N)
Mast tilt towards radiometer (- if tilting towards anemometer)	+6 °	+2 °
Mast tilt across boom looking from radiometer (- if clockwise)	* ₂ -2 °	0 °
Boom direction relative to north	0 ° true/magn.	0 ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	(Y/N)	(Y/N)
Radiometer exactly aligned with inclinometer? If not, measure.	(Y/N)	(Y/N)
Temperature / humidity height (bottom of casing – surface)	264 cm	237 cm
Stake assembly standing? But connector-stake is sloppy	(Y/N / sort of)	(Y/N / sort of)
Free length of stake (if 3: outer, holding boom with sonic ranger)	229 cm	225 cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	188 cm	160 cm
Vertical difference in surface height (at station – at hose) (approx.)	30 cm	30 cm
Length of thermistor string on surface from surface marking	85 cm	85 cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

- Ablation nose falling into crevasse
- Data would not download. Got error message (see photo documentation) saying: duplicate entries. Took card, and ~~replaced~~
- Had trouble connecting to the logger $\Rightarrow$ several attempts before success with both loggers. Issue with tough book or cable?

Must-do at next visit:

- velcro for new GPS
- no velcro on modem: put velcro

- *₂ Boom was tilted, screws were loose.
- * lost phone with photos.

After service
 SR50 on free stake: 130 cm
 SR50 on station: 258

- When uploading new logger-program it seemed to fail or never finish. But after restarting logger-program, it was installed.

Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	TAS SF
People present	ASO + ARU
Purpose	Maintenance / Installation / take-down / quick visit / emergency
Date	10 May 2023

Notes:

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N				Clean? Y / N
Ablation hose assembly	N				No air in hose and bladder half full? Y / N
Iridium modem IMEI number	N				
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	N				
Wind sensor	N		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	N		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	N		Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	N		Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	N				
Solar panel	N		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N		At top of logger enclosure? Y / N		
Logger box ID	N		CR1000 or CR1000X? ID number:		
Multiplexer	N		Clicking during measurement cycle? Y / N		
Logger enclosure	N		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y	N	New box? Y / N New desiccant bags? Y / N		
Stakes	N		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N		New tripod? Y / N New sensor boom? Y / N		

Station name	TAS SF
Date	10 aug 2023

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	Y / N
Temperature / humidity height (bottom of casing – surface)	cm	cm
Stake assembly standing?	Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y / N	Y / N

Notes:

- * Cable from solarpanel ripped out of plug. Did not have time to repair.
- * Took old battery box home. Did not install new battery.
- * Did not download data.

Must-do at next visit:

- * New battery
- * New solar panel.
- ✓ Get data.